IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

C++ Systems & Software Engineering

An 8–Week Professional Internship Presentation Report

INTERNSHIP DURATION
June 2026 – August 2026 (8 Weeks)

STUDENT CREDENTIALS

Soham Nanda

ENROLLMENT / ROLL NO.
25SCS1003001673

PROGRAM & SEMESTER
B.Tech (CSE)

HOST ORGANIZATION
EduSkills Academy
Supported by AICTE National Internship Portal

PROFILE SUMMARY

Candidate & Program Overview

Candidate Details

Student Name:	Soham Nanda
Roll Number:	25SCS1003001673
Degree Track:	B.Tech (CSE)
Batch & Term:	2026–2027
University:	IILM University, Greater Noida

Internship Profile

Program:	AICTE – EduSkills Virtual
Domain:	C++ Systems Programming
Duration:	8 Weeks (June – August 2026)
Host Partner:	EduSkills Academy
Focus Standards:	Modern C++17 / C++20 Core

HOST ORGANIZATION

Organization Profile: EduSkills Academy

ISO 9001:2015 ISO 20000-1:2018

EduSkills operates with a national mandate for "Nation Building Through Skills," bridging academia with high-impact software engineering competencies.

AICTE Patronage

National portal integration for certified skill readiness.

Industry Alignment

Curriculum built to meet enterprise systems benchmarks.

Core Tech Focus

Systems engineering, modern C++, algorithms, and low-latency.

DOMAIN RELEVANCE

Problem Statement & Industry Context

The Industry Challenge

Standard academic coursework frequently limits C++ to basic syntax and procedural assignments, omitting modern systems paradigms.

- ✕ Severe memory leaks via raw pointers.
- ✕ Zero exposure to multithreaded race conditions.
- ✕ Lack of automated profiling and build systems.

The Practical Solution

An immersive technical pipeline focusing on modern C++17/20 practices, RAII safety, concurrent routines, and native system optimization.

- ✓ Deterministic RAII and smart pointer safety.
- ✓ Low-latency data structures via STL algorithms.
- ✓ High-throughput thread pools and mutex locks.

INTERNSHIP TARGETS

Core Internship Objectives

RAII & Memory

Zero-leak architectures using smart pointers.

Advanced OOP

Compile-time polymorphism and meta-templates.

STL Mastery

Optimal container selections and custom iterators.

Concurrency

Thread synchronization and lock-free routines.

TECHNICAL STACK

Modern C++ Systems Toolchain

Languages

Modern Standards: C++17, C++20

Low-Level: C compatibility

Compilers

Toolchains: GCC (g++), Clang/LLVM

Build Engine: CMake, GNU Make

Debugging

Diagnostics: GDB, LLDB

Profilers: Valgrind, Sanitizers

Libraries

Standard: STL Containers & Algorithms

Testing: Google Test (GTest)

ARCHITECTURE & DESIGN

C++ High-Performance Systems Architecture

APPLICATION LAYER	High-performance command-line engines, interactive simulators, and system services implementing strict domain models.
CONCURRENCY	Multithreading workers, thread-safe queues, atomic primitives, and non-blocking task schedulers via and .
STL & TEMPLATES	Generic template metaprogramming, lambda functions, and vectorized operations over associative and sequence containers.
MEMORY & RAII	Custom allocators, explicit move semantics (`std::move`), and scoped smart pointers (`unique_ptr`, `shared_ptr`).

WEEKLY PROGRESSION

Curriculum Timeline (Weeks 1 to 4)

WEEK 1

C++ Fundamentals

Data types, bitwise operators, pointers, and memory layout.

WEEK 2

Object-Oriented C++

Class inheritance, vtables, encapsulation, and virtual methods.

WEEK 3

Memory & Pointers

Dynamic heap allocation, move semantics, and RAII safety.

WEEK 4

STL & Generics

Vector, map, custom templates, and algorithm pipelines.

WEEKLY PROGRESSION

Curriculum Timeline (Weeks 5 to 8)

WEEK 5

Modern C++ Standards

C++17 structured bindings, optionals, and lambdas.

WEEK 6

Multithreading

Threads, mutex locks, and concurrent worker queues.

WEEK 7

Build & Optimization

CMake configuration, GDB debugging, and Valgrind.

WEEK 8

Capstone Delivery

Production system project and unit test benchmarks.

INTERNSHIP DELIVERABLES

Key Projects & Demonstrated Skills

Custom Smart Pointer & Memory Pool Allocator

Built RAII-compliant ref-counted smart pointers and custom block memory managers.

High-Throughput Concurrent Thread Pool

Engineered task-stealing worker threads with mutex locks and condition variables.

Low-Latency Order Book & Data Indexing Engine

Developed fast search structures leveraging STL unordered maps and cache-aligned arrays.

Automated Diagnostic & Leak Detection Suite

Configured CI pipelines with Valgrind and Google Test for automated memory verification.

Thank You!

Soham Nanda

Roll No: 25SCS1003001673 | B.Tech (Computer Science & Engineering)

School of Computer Science & Engineering

IILM University, Greater Noida, Uttar Pradesh